Ayesha Firdaus

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PROFESSIONAL SUMMARY

Data Science and Analytics professional with a biopharma operations background, skilled in Python, SQL, Power BI, and ML with working knowledge of NLP techniques (traditional and deep learning). Built four end-to-end projects spanning risk detection, customer analytics, forecasting, and optimization, each with quantified outcomes and stakeholder-facing dashboards

SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn, Matplotlib), SQL (SQL Server, BigQuery)

ML & Modeling: LightGBM, XGBoost, Regression, Classification, Hypothesis Testing, Feature Engineering

NLP & Text: Traditional NLP (tokenization, TF-IDF, text classification), Deep Learning NLP (transformers, embeddings)

BI & Reporting: Power BI, Excel (advanced), Looker Studio, Metabase

Data & Tools: EDA, ETL/ELT, Data Cleaning, Data Modeling, Git, Google BigQuery

WORK EXPERIENCE

MSAT Intern — 09/2024 – 05/2025 | Bengaluru, India
Syngene International
- Monitored and analysed batch manufacturing data across multiple client projects; tracked parameter distributions, flagged outliers, and reported findings to manufacturing and stakeholder teams — corrective actions improved batch purity from ~90% to 98% on a key protein project
- Built Excel-based analysis workflows for parameter tracking across batches: plotted distributions, identified process deviations, and supported root cause investigations for out-of-range values
- Coordinated with QC teams on documentation covering raw materials, consumables, and vendor specifications; validated inputs, reconciled outputs, and closed documentation loops
- Worked across projects for clients including BMS and Zoetis; managed documentation, cross-functional coordination, and ad hoc analytical requests in parallel

PROJECTS

Banking Fraud Detection & Transaction Risk Analytics — 03/2026 – 04/2026
SQL Server, Python, XGBoost, Power BI
- Designed a four-layer fraud detection system: SQL analytics, behavioral risk scoring, XGBoost (ROC-AUC 0.844), and Power BI dashboards across 44K+ transactions; identified $380M in fraud exposure
- Custom risk engine stratified transactions into four bands; Critical Risk segment isolated at 60%+ fraud rate, flagging 665 high-priority customers for investigation
- Key fraud signals: crypto merchants (40.1% fraud rate), 1AM peak transactions (23.4% fraud rate), and fraud occurring ~6× farther from customer home locations surfaced via 4-page executive dashboard

Customer Channel Affinity Prediction System — 04/2026 – 05/2026
Python, LightGBM, Streamlit
- Developed a multiclass ML model using 140+ engineered behavioral features across 9 data sources to predict future customer channel affinity
- Implemented leakage-safe temporal validation; LightGBM achieved 33.5% top-1 / 70.4% top-3 accuracy (2.3× lift over 14.3% random baseline)
- Deployed full pipeline as a live Streamlit app with customer-level channel recommendations, probability rankings, and budget allocation simulator

Dynamic Pricing & Demand Forecasting System — 03/2026 – 05/2026
Python, LightGBM, Streamlit
- Built an end-to-end ML pipeline forecasting hourly demand and computing revenue-optimal dynamic prices
- Engineered 23 features on 8,737 hourly records using lag windows, DSR, and event flags
- Achieved 3.1% MAPE vs 16.7% naive baseline, delivering +12.7% revenue uplift (₹1.45M annualised) with 83.7% demand retention

Supply Chain Inventory Optimization & Demand Forecasting — 04/2026 – 05/2026
SQL Server, Python, Power BI
- Built an end-to-end analytics pipeline diagnosing inventory inefficiencies across 6 supply chain tables; engineered reorder points, safety stock, and turnover KPIs in SQL; built 30-day rolling demand forecasts in Python
- Identified ~95% of products were overstocked, total inventory held at ~23× forecasted monthly demand; surfaced via a 3-page executive Power BI dashboard with product-level drill-down and supplier delay analysis
- Flagged Electronics as the lowest turnover, highest-priority category for inventory reduction; quantified a potential reduction of ~196 units per product against current avg stock of ~221

EDUCATION

Vellore Institute of Technology — 09/2020 – 08/2025 | Vellore, India
MSc. Integrated Biotechnology
CGPA - 8.31/10
Relevant coursework: Statistics, Data Analysis, Research Methodology

AnalytixLabs & E&ICT Academy, IIT Guwahati — 08/2025 – Present | Bengaluru
Advanced Certification in Data Science (ACDS)